Introduction to Limits

ANSWER KEY

Evaluating limits

1 Evaluate $\lim_{x\to 2}\frac{x^2-4}{x-2}$ by first simplifying the expression.
$\frac{(x-2)(x+2)}{x-2} = x+2$ for $x \neq 2$, so the limit is 4.

2 Evaluate by direct substitution where possible:

a $\lim_{x\to 3}(x^2+2x-1) \quad = 14$

b $\lim_{x\to 1}\frac{x+5}{x+1} \quad = 3$

3 Using a table of values near $x=0$ (in radians), estimate $\lim_{x\to 0}\frac{\sin x}{x}$.
Values approach 1 from both sides; the limit is 1.

One-sided limits and limits at infinity

4 Let $f(x)=\frac{|x|}{x}$. Find:

a $\lim_{x\to 0^+} f(x) \quad = 1$

b $\lim_{x\to 0^-} f(x) \quad = -1$

c $\lim_{x\to 0} f(x)$ does not exist

5 Evaluate each limit at infinity:

a $\lim_{x\to\infty}\frac{3x^2+5}{x^2-2} \quad = 3$

b $\lim_{x\to\infty}\frac{2x+1}{x^2+4} \quad = 0$

6 The graph of $f(x)=\frac{1}{x-2}+3$ has asymptotes $x=2$ and $y=3$. Express each asymptote as a limit statement.
$\lim_{x\to 2^+} f(x) = +\infty$ (and $\lim_{x\to 2^-} f(x) = -\infty$); $\lim_{x\to\pm\infty} f(x)=3$.